Population Study of Bangladesh

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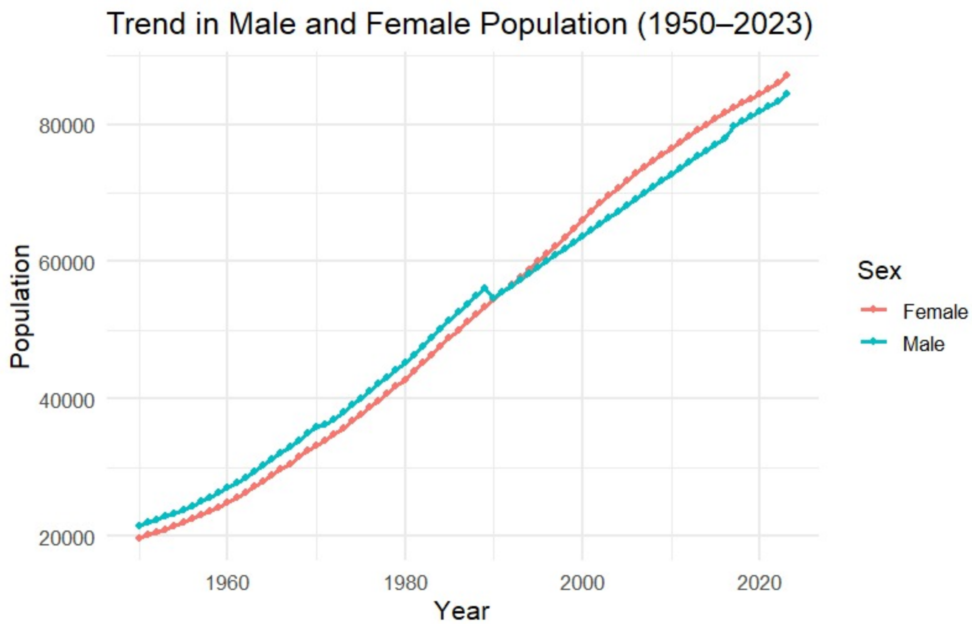
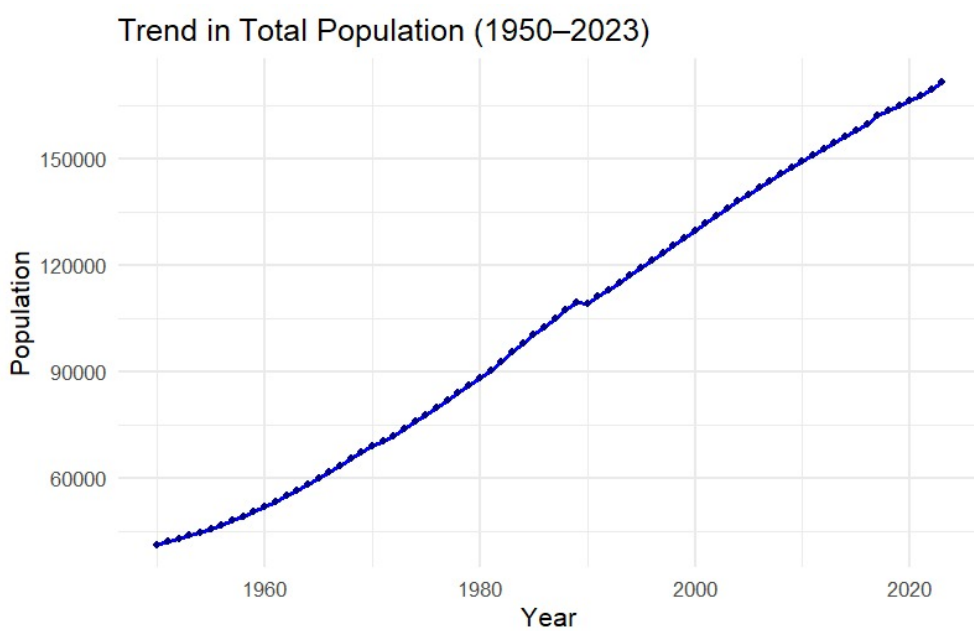
1 Introduction

Population dynamics play a crucial role in shaping the economic, social, and developmental trajectory of any nation. By studying a country's growth trends, dependency ratios, and demographic patterns, we can gain a deeper understanding of the opportunities and challenges it faces. This project focuses on analyzing the population trends of Bangladesh, evaluating its year-on-year growth, demographic dividend, and dependency ratios in comparison with global averages. Furthermore, the study examines how historical data on resources and crop yields can be linked with classical population theories such as Malthusian, Boserup, and the Demographic Transition Theory (DTT). Through this, we aim to assess the country's stage in demographic transition and its implications for future development. Furthermore, we also made a comparison study on the population pyramid of Bangladesh over the years 1951-2023.

2 Plots


Trend in Male and Female Population (1950–2023)
Population
80000
60000
40000
20000
1960
1980
2000
2020
Year
Sex
Female
Male


Trend in Total Population (1950–2023)
Population
150000
120000
90000
60000
1960
1980
2000
2020
Year

3 Population pyramid over the Years 1951-2023 and its transition

Link:https://github.com/ANJAN25032025/population-project
The animation of the population pyramid across years reveals how the demographic structure of a country evolves over time.

Early Years (Expansive Pyramid)

- The base of the pyramid is wide, indicating **high birth rates**.
- A large proportion of the population is concentrated in the younger age groups (0–14 years).
- This suggests limited healthcare, high fertility, and possibly higher infant mortality.

Middle Years (Transitional Pyramid)

- The base begins to **narrow**, showing that birth rates are declining.
- The middle of the pyramid (working-age population, 15–64 years) **expands**, reflecting improvements in healthcare, education, and economic opportunities.
- This stage often corresponds to the **demographic dividend**, when the working-age population outnumbers dependents, boosting economic growth potential.

Recent Years (Constrictive Pyramid)

- The top of the pyramid widens, representing **increasing life expectancy** and a growing elderly population (65+ years).
- The base becomes relatively narrow, showing **low fertility rates** and slower population growth.
- This signals an **aging population**, which can create challenges such as increased demand for healthcare, pensions, and social support.

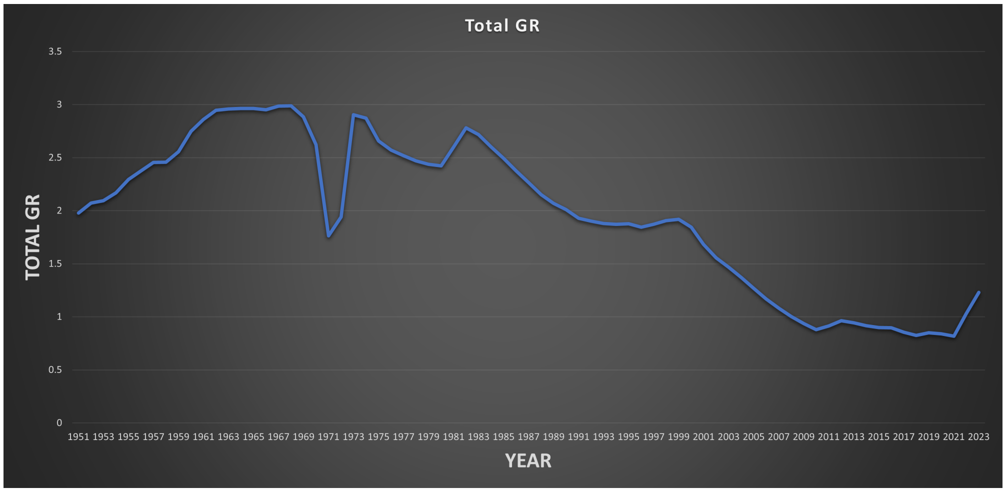
Overall Trend

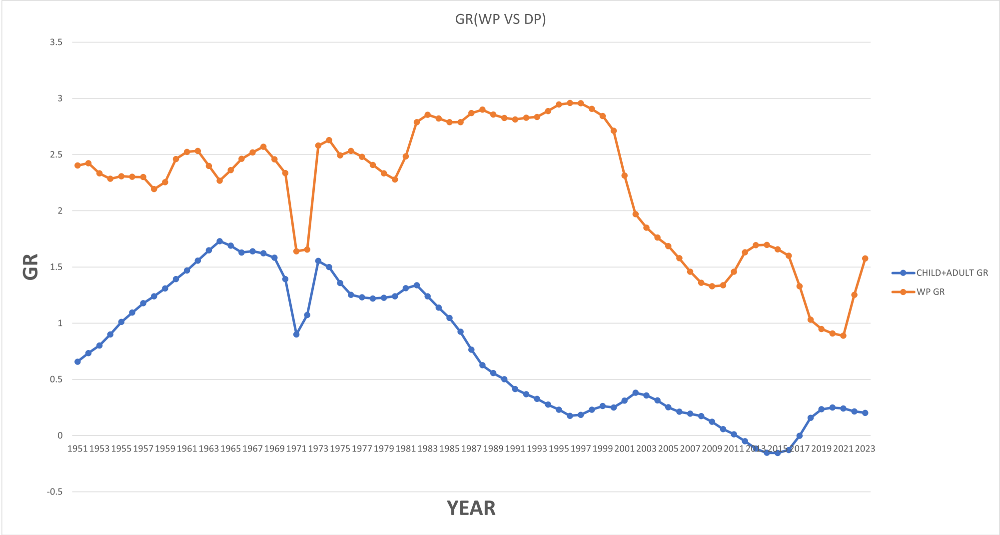
The transition from a wide-based to a narrow-based pyramid reflects the **demographic transition model**:

1. **Stage 1–2:** High fertility and mortality.
2. **Stage 3:** Falling fertility, rising life expectancy.
3. **Stage 4–5:** Aging population, low fertility, possible population decline.

4 Year-wise Description of Growth Rates in Bangladesh (1951-2023)

4.1 Plots





4.2 Interpretation of the plot for working age population growth rates vs dependents(child+adult) growth rates

1950s–1960s

- **Child + Adult Dependent Growth Rate (blue):** Rising steadily, peaking around 1964–65 at nearly 1.7%. This shows a rapid increase in population, especially among children, consistent with high fertility and declining infant mortality after independence from British India.
- **Working Population Growth Rate (orange):** Stable around 2.3–2.5%, reflecting a young but expanding labor force.

1970–1975

- **1971 Liberation War:** Both WP and DP growth rates plummeted sharply around 1971–72 due to large-scale deaths, displacement, and destruction during the Bangladesh Liberation War.
  - DP GR (blue) dropped close to 0.9%, while WP GR (orange) also dipped to around 1.6%.
- **1974 Famine:** Population growth suffered another setback in the mid-1970s with widespread famine. This explains the fluctuations and slower recovery in both dependent and working population growth during 1973–75.

Late 1970s–1980s

- **Dependents (blue):** After recovering from the famine, the dependent population growth stabilized around 1.2–1.4%, but started declining after 1985.
- **Working Population (orange):** Increased strongly to nearly 2.8–2.9% by the mid-1980s, reflecting the demographic momentum.
- This period marks the early demographic transition—fertility decline begins, but the working population expands rapidly.

1990s

- **Dependents:** Show continuous decline, falling below 0.5% by 1995–2000, signaling fertility reduction and smaller family sizes.
- **Working Population:** Peaks near 3.0% in the early 1990s, creating a demographic dividend opportunity.
- **1999–2000 (Sierra Leone War):** Bangladesh participated in UN peacekeeping operations in Sierra Leone during this period. The domestic demographic situation showed:
  - Low dependent growth (near zero).
  - Working population still strong ($\approx 2.5\%$).

  This reflects Bangladesh's growing role in the international community, supported by a robust labor force.

2000s

- **Dependents:** Growth rate becomes nearly stagnant (~ 0.2–0.3%), showing population stabilization in child cohorts.
- **Working Population:** Declines gradually from $\sim 2.5\%$ (2000) to 1.5% (2010), reflecting slowing labor force growth as fertility drops further.

2010s

- **Dependents:** Near zero or negative growth around 2012–2016, indicating aging population trends and very low fertility.
- **Working Population:** Continues to decline, dipping below 1.0% by 2017–18.
- **2019 Dengue Outbreak:** A major dengue outbreak in Bangladesh coincides with the period when both WP and DP growth rates were already low. While not directly visible in macro growth rates, it reflects the public health challenges faced during a slowing demographic momentum.

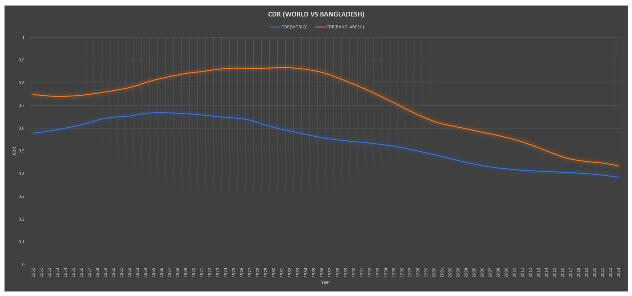
2020–2023

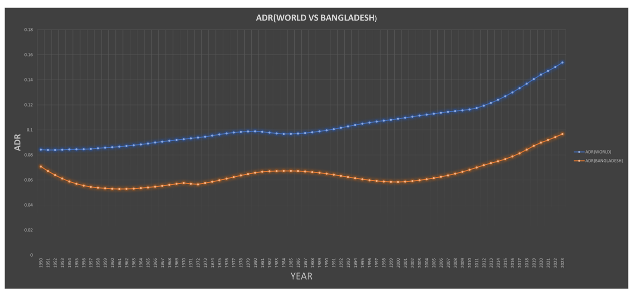
- **COVID-19 Pandemic and Recovery:** Both series show a slight rebound after 2020.
- **Dependents:** Growth rate rises modestly (≈ 0.3–0.4%), possibly due to short-term fertility and mortality shifts.
- **Working Population:** Recovers from its lowest point ($< 1\%$ in 2019) to $\sim 1.6\%$ in 2023, suggesting a temporary demographic boost post-pandemic.
- **Migration Effects:** Bangladesh has a high rate of international labor migration (Middle East, Southeast Asia). During COVID, many migrants returned home, boosting the domestic working population temporarily. As migration resumed in 2021–22, the labor force remained large but more stable.

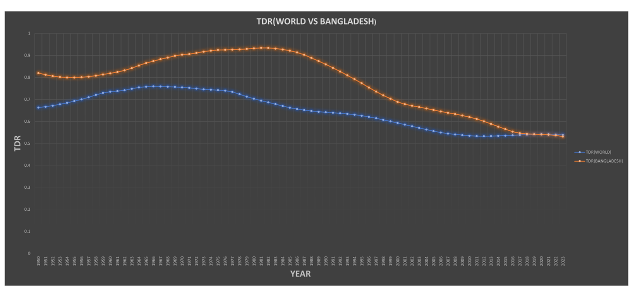
Bangladesh vs World: Dependency Ratios (CDR, ADR, TDR)

What the ratios mean.

- **Child Dependency Ratio (CDR)**: dependents aged 0–14 per 100 people aged 15–64.
- **Aged (Old-age) Dependency Ratio (ADR)**: dependents aged 65+ per 100 people aged 15–64.
- **Total Dependency Ratio (TDR)**: CDR + ADR; the overall burden on the working-age population.







How to read the chart.

- Each line shows the ratio through time. A *declining CDR* usually signals falling fertility and progress through the demographic transition.
- A *rising ADR* indicates population ageing; this can lift the TDR even when the CDR is falling.
- The *gap between Bangladesh and the World* tells you whether Bangladesh's pressure on workers is above or below the global average, and whether it is converging.

Key comparison (typical pattern seen in the graph).

- **CDR:** Bangladesh generally starts above the World average (younger age structure) and *declines faster*, shrinking the gap over time. This reflects sustained fertility decline.

- **ADR:** Bangladesh's ADR is typically *below* the World average for many years but *rises steadily* as longevity improves and cohorts age. The gap tends to narrow in the most recent years.
- **TDR:** With CDR falling faster than ADR rises, Bangladesh's TDR shows a clear *downward trend*, indicating a lighter overall burden on workers and an extended *demographic dividend* window. In recent years, TDR approaches (or converges toward) the World average.

Implication. Bangladesh's rapid CDR decline has expanded its working-age share, supporting growth and savings. However, the gradual rise in ADR means policies for healthy ageing, pensions, and productivity gains will matter more going forward, so that TDR remains manageable as the population structure matures.